Nearest Neighbor Reasoning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected nearest-neighbour record.

Measure distances, apply k/ties, preserve split independence and limit vote/probability claims.

Keep one habit: Retain training rows, query values, selected IDs, predictions, references and baseline separately.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Similarity by Selected Traits

Sampling and Representation

Overfitting Intuition

W1 · Calculate all five feature distances

> Labels do not affect the numeric gap.

Your evidence record

- Worked training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear; unitscm. Distance is absolute difference, distance ties sort alphabetically by ID. Compare k1/k3; k3 votes by majority of stored labels. Worked validation queries/references:(3,blocked),(9,blocked),(11,clear). The baseline is training-majority clear. The separate compact/extended book pool has different rows and is not used for this validation.

- For query 3 cm, calculate **distance**s to all five **training** rows. Explain why A and B tie even though their labels differ.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Apply k and the declared distance tie-break

> Changing the number of voters can change the prediction.

Your evidence record

- Worked training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear; unitscm. Distance is absolute difference, distance ties sort alphabetically by ID. Compare k1/k3; k3 votes by majority of stored labels. Worked validation queries/references:(3,blocked),(9,blocked),(11,clear). The baseline is training-majority clear. The separate compact/extended book pool has different rows and is not used for this validation.

- For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Keep query answers out of the pool

> Use all three supplied validation records and the same baseline.

Your evidence record

- Worked training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear; unitscm. Distance is absolute difference, distance ties sort alphabetically by ID. Compare k1/k3; k3 votes by majority of stored labels. Worked validation queries/references:(3,blocked),(9,blocked),(11,clear). The baseline is training-majority clear. The separate compact/extended book pool has different rows and is not used for this validation.

- Why must the three validation reference labels not enter the **neighbour** pool? Give the fixed k=3 score and baseline score on the worked validation examples.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit a vote fraction and an exact match

> Neither establishes calibrated correctness or full suitability.

Your evidence record

- Worked training A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear; unitscm. Distance is absolute difference, distance ties sort alphabetically by ID. Compare k1/k3; k3 votes by majority of stored labels. Worked validation queries/references:(3,blocked),(9,blocked),(11,clear). The baseline is training-majority clear. The separate compact/extended book pool has different rows and is not used for this validation.

- Two of three **neighbour**s **vote** clear. Does that prove a two-thirds chance of a correct **prediction**? Does an exact feature match guarantee a correct label? Explain both limits.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Find neighbours independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use the new P–T training pool supplied in W4. Distance is absolute gap, equal distances use alphabetical ID order and k3 votes use the three stored labels; do not add validation references to training.

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Find neighbours independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Use **neighbor**ing liked books as **evidence**, while explaining which traits created closeness.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Apply **neighbour** and **tie rule**s for Nearest **neighbour**s and **distance**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A query has nearest books A and B at **distance** 1, with conflicting labels; C is **distance** 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Find nearest **neighbour**s using differing-trait **distance** and k=1.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Training items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute **distance** to choose its nearest **neighbour**.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 22 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A query has nearest books A and B at **distance** 1, with conflicting labels; C is **distance** 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

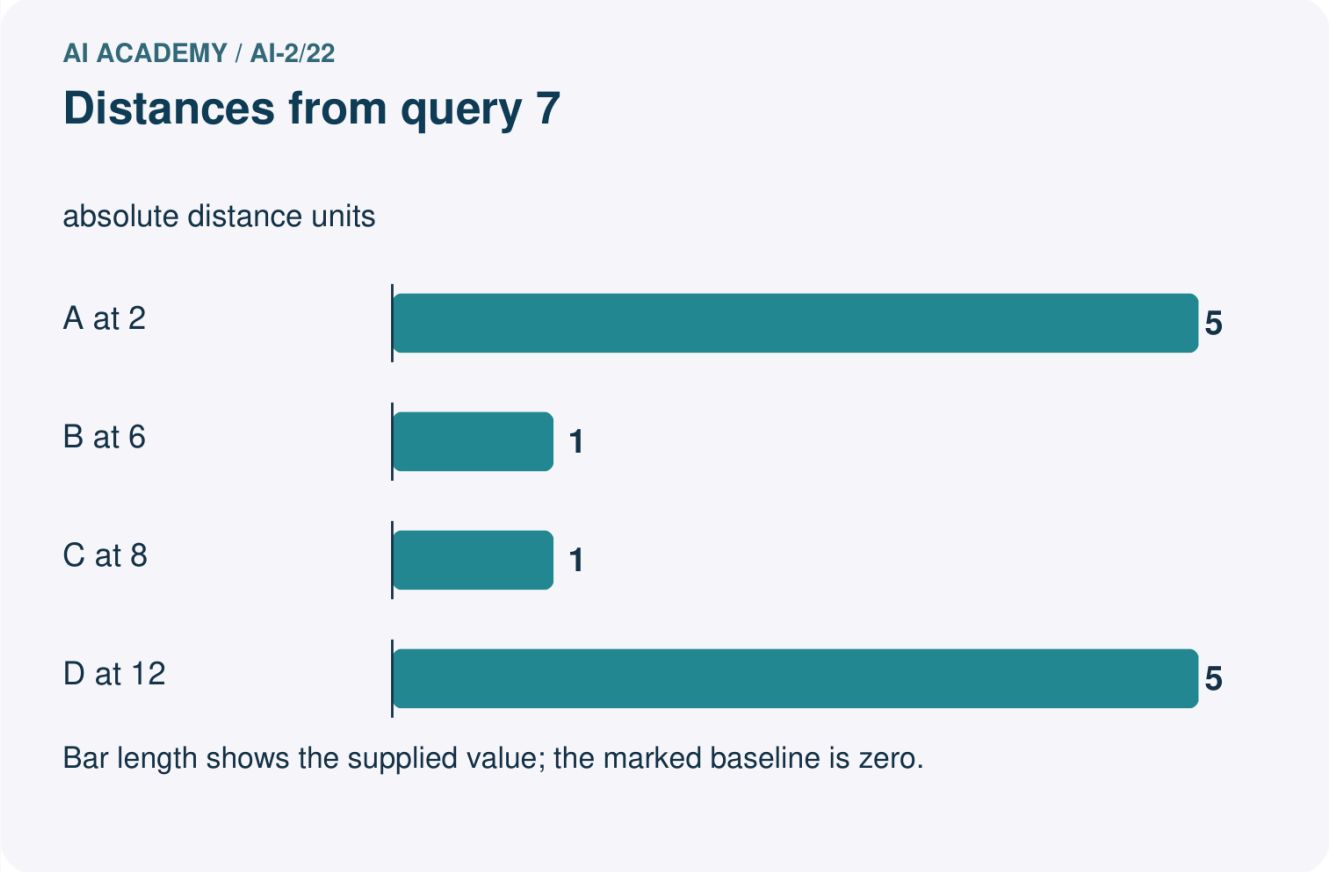
> Keep the explanation and the support you used.

Use this when helpful

A fictional one-feature classifier predicts compact or extended book format.

Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/22

Nearest Neighbor Reasoning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

From 7: A=5, B=1, C=1, D=5.



2. Order neighbours

Distance then ID gives B,C,A,D.



3. k=1 prediction

First neighbour B is extended: predict extended.



4. k=3 prediction

B,C,A vote extended,compact,compact: predict compact.

CHANGE ONE THING

Move query from 7 to 9 while keeping training fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why do k=1 and k=3 disagree for the original query?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional one-feature classifier predicts compact or extended book format.
- Which labelled step matches this detail? B,C,A **vote** extended,compact,compact: predict compact.
- 1. k=1 **prediction**
- 2. Order **neighbour**s
- 3. k=3 **prediction**
- 4. Measure **distance**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: k=3 **prediction** B,C,A **vote** extended,compact,compact: predict compact. Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed.

The query's reference label must not enter the stored pool before predicting. Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Changed-case question

- Changed-case practice: Move query from 7 to 9 while keeping **training** fixed. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Repair and save your work

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Nearest Neighbor Reasoning: the case

AI ACADEMY / AI-2/22 · 01 · LOOK AT THE FACTS

Nearest Neighbor Reasoning: the case

The situation

A fictional one-feature classifier predicts compact or extended book format.

Supplied evidence

Stored training rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.
Distance is absolute difference; ties sort by ID; tied category vote chooses compact.

By the end, I can order training neighbours, apply k and keep query labels out of prediction.

Follow the labelled picture with the matching explanation. Open the vector diagram

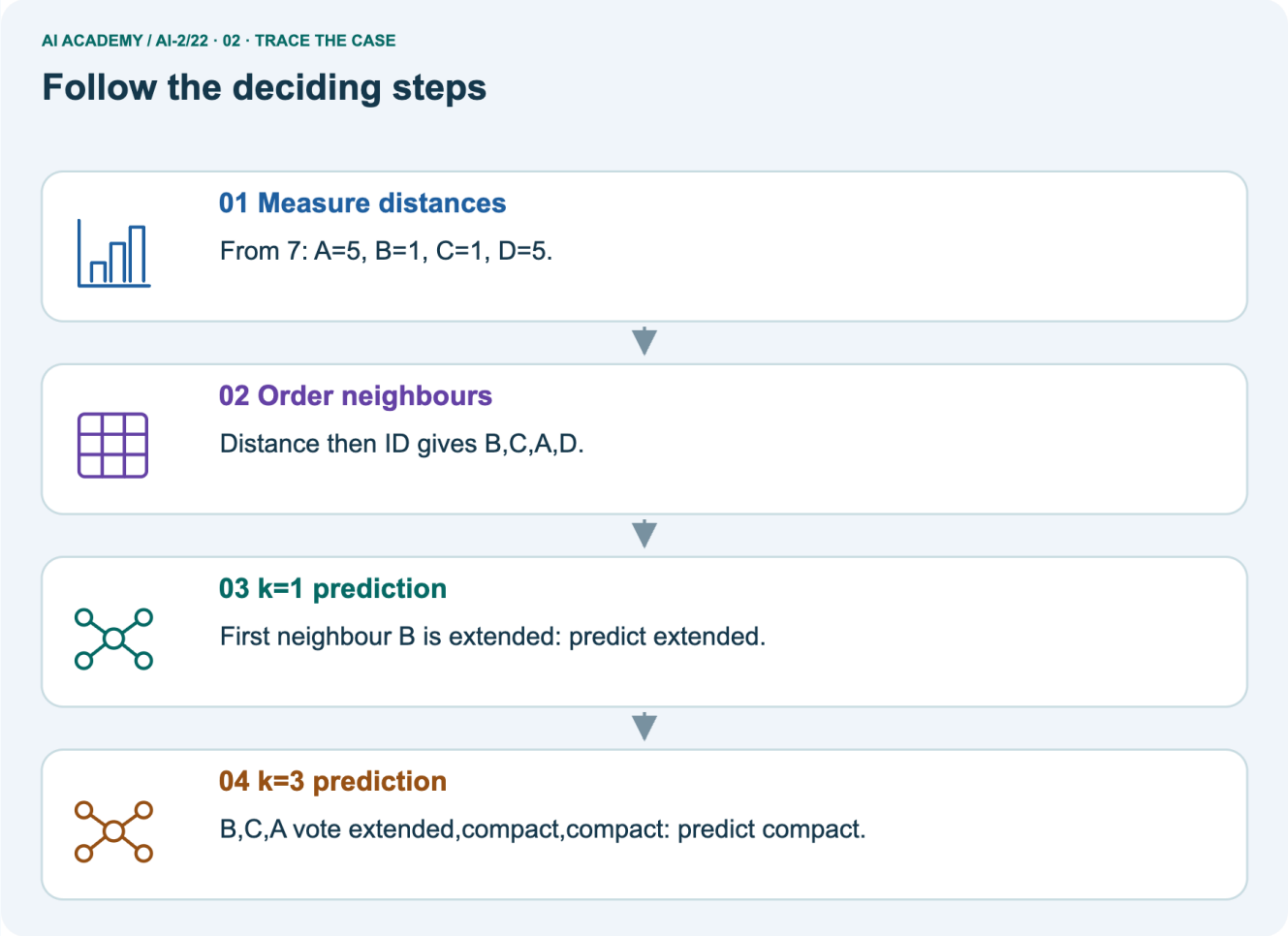
A fictional one-feature classifier predicts compact or extended book format. Supplied facts: Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



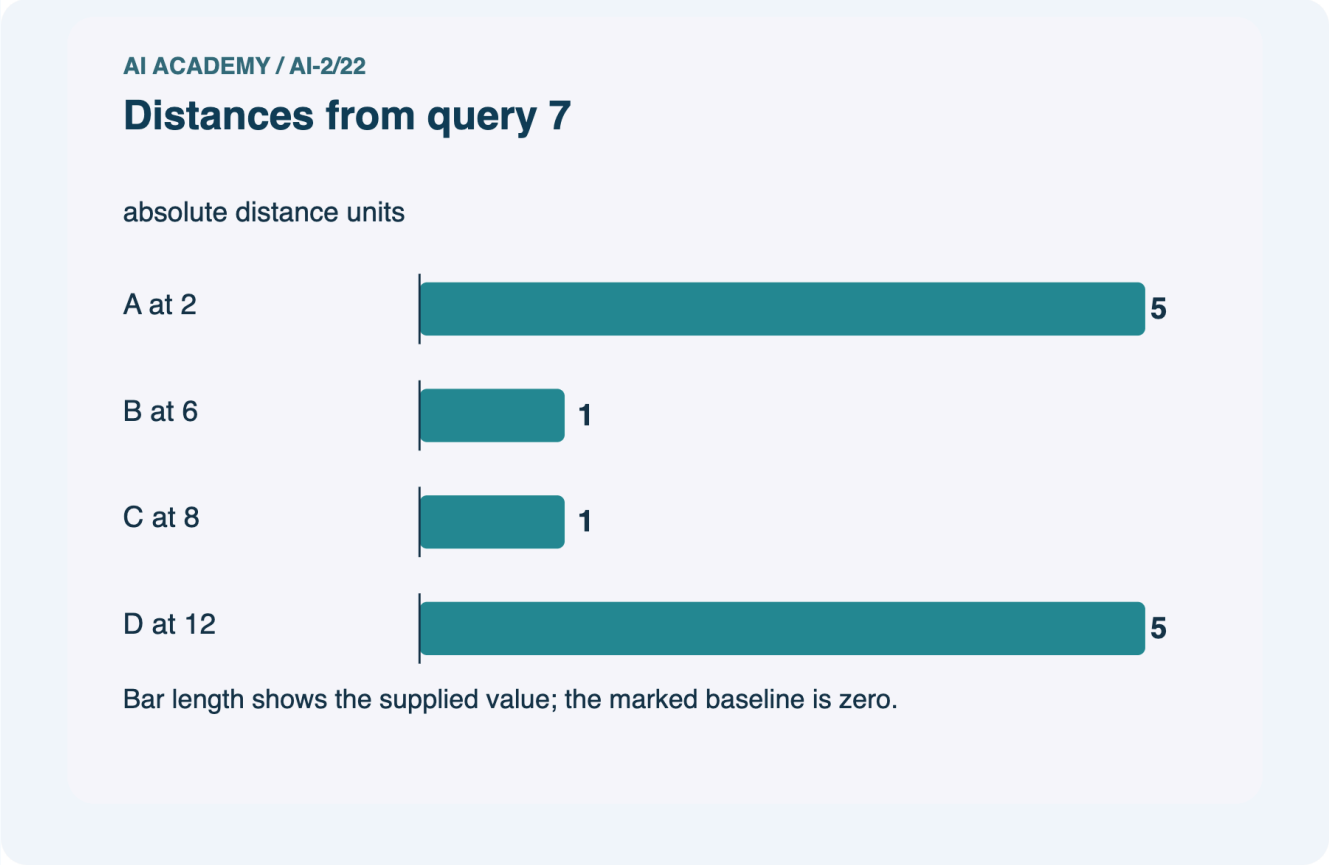
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: From 7: A=5, B=1, C=1, D=5. → Order **neighbour**s: **Distance** then ID gives B,C,A,D. → k=1 **prediction**: First **neighbour** B is extended: predict extended. → k=3 **prediction**: B,C,A **vote** extended,compact,compact: predict compact.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Distances from query 7



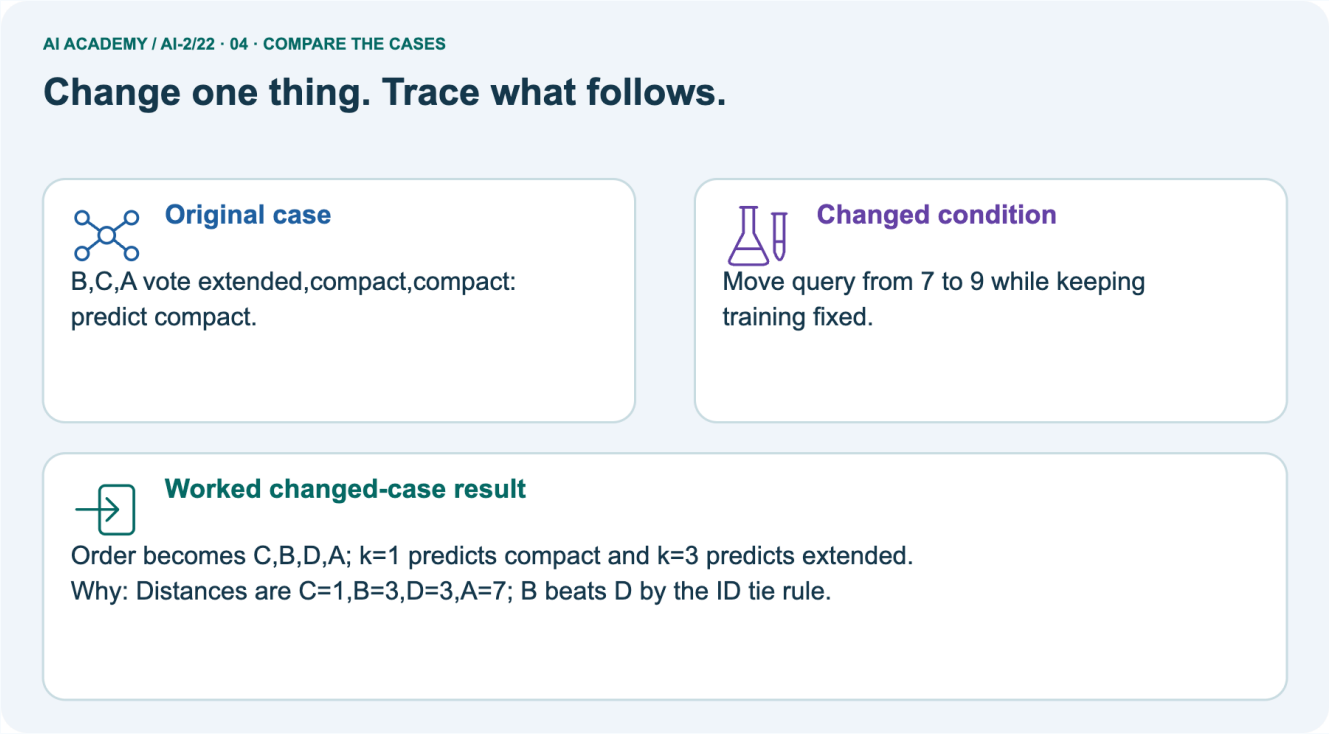
Follow the labelled picture with the matching explanation. Open the vector diagram

Distances from query 7. Units: absolute **distance** units. A at 2: 5. B at 6: 1. C at 8: 1. D at 12: 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

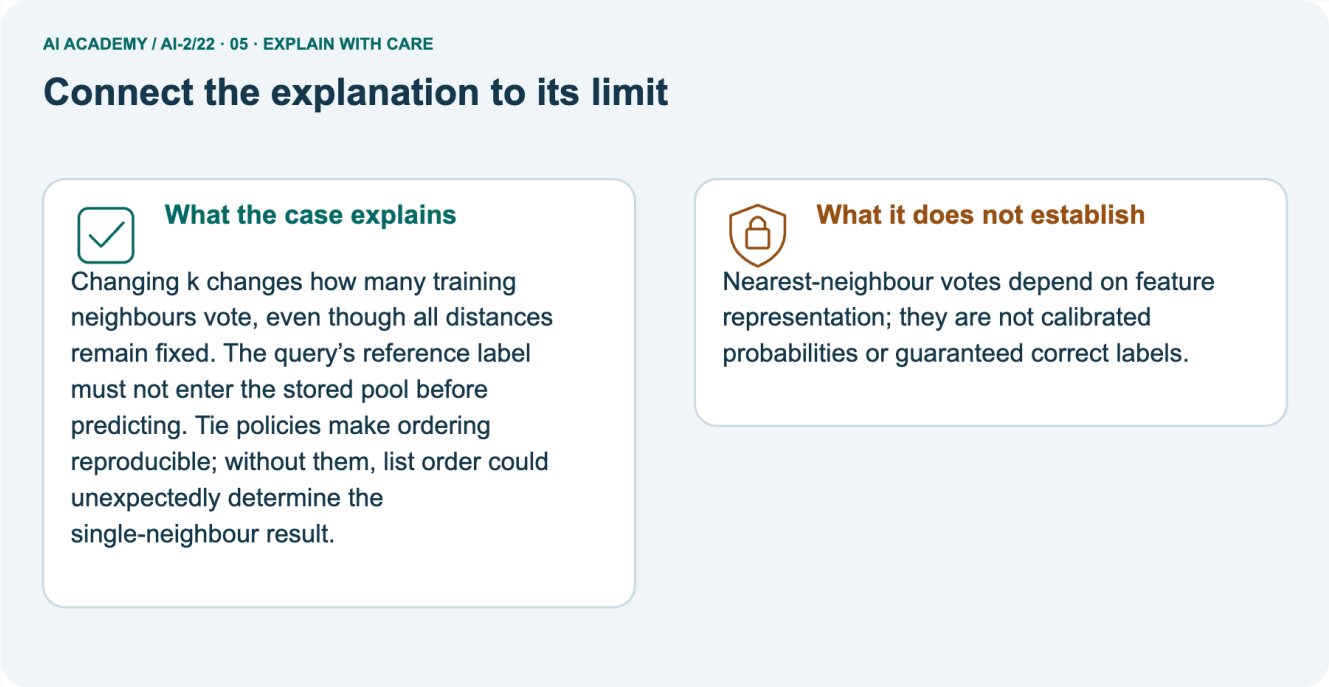
Original case: B,C,A **vote** extended,compact,compact: predict compact. Changed condition: Move query from 7 to 9 while keeping **training** fixed.

Expected result: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

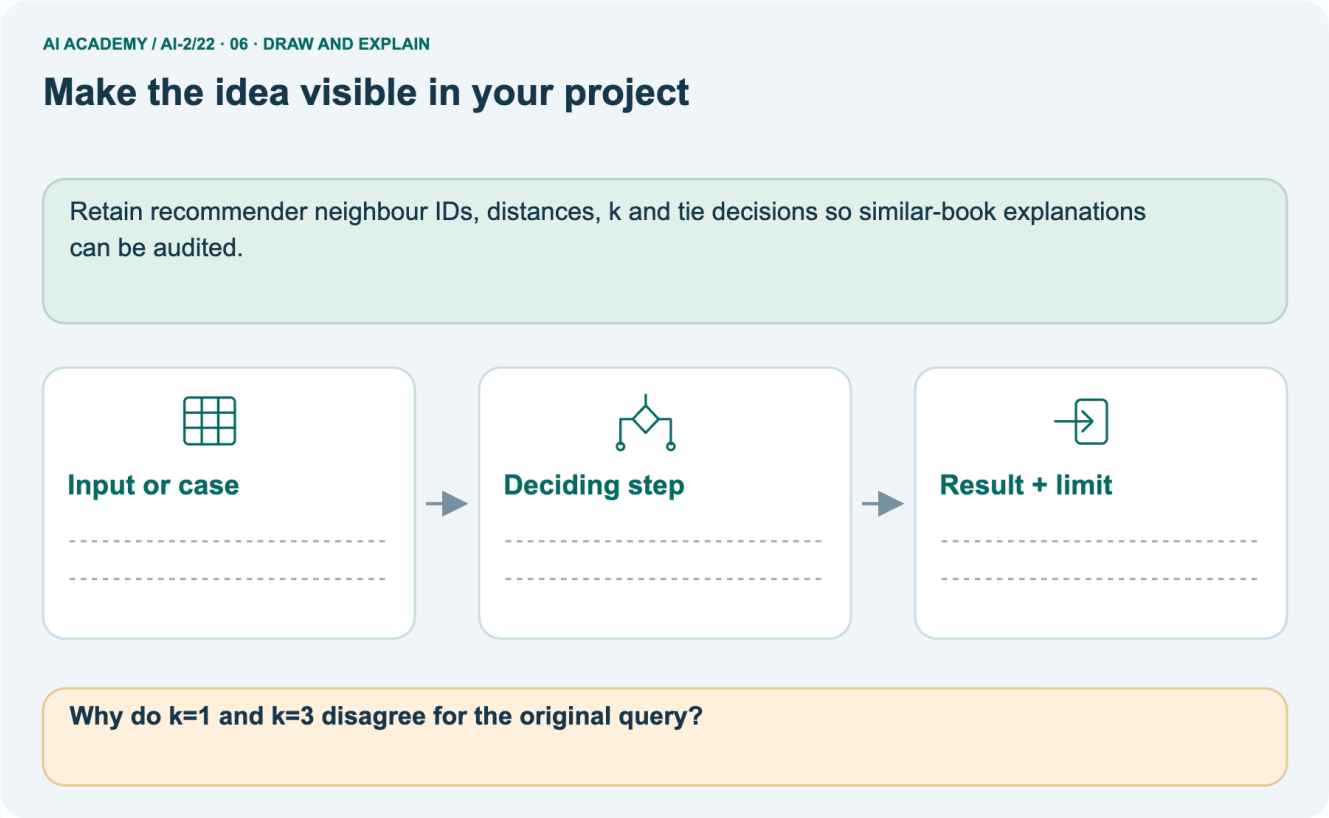
Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Limit: Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited. Draw your **input**, deciding step, result and limit.

Supported question: Why do k=1 and k=3 disagree for the original query?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Nearest Neighbor Reasoning: the case

02. Follow the deciding steps

03. Distances from query 7

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.